AQ9.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Partial Derivatives:

Level 1:

What is the rate of change of $f(x, y, z) = 5x^2 - 6xy + 3z^2 - 2yz$ at (1,2,-1) with respect to z?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -10
1 point

1 point
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as $f(x, y) = x^3 + 3xy$. Identify the correct option with respect to the partial derivatives of f.

- [ ] $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$
- [ ] $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0)$
- [ ] $\frac{\partial f}{\partial x}(1, 1) = \frac{\partial f}{\partial y}(1, 1)$
- [ ] $\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0)$
- [ ] $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} + 3$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0)$
$\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0)$

1 point
Let $f(x, y) = \frac{xy}{x^2 + y}$. Identify the correct options with respect to the partial derivatives of f.

- [ ] $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not defined at $(0, 0)$
- [ ] $\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 1/4$
- [ ] $\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 0$
- [ ] $\frac{\partial f}{\partial x}(1, 0) = 0, \frac{\partial f}{\partial y}(1, 0) = 1$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not defined at $(0, 0)$

$\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 1/4$

$\frac{\partial f}{\partial x}(1, 0) = 0, \frac{\partial f}{\partial y}(1, 0) = 1$

Let $f: R^2 \rightarrow R$ be defined as $f(x, y) = x^2 + 4xy - y^2$. Suppose $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at $(12, k)$. Find the value of k

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

Level 2:

1 point
Suppose $f(x, y, z) = xy + yz + zx - 3xyz$ is a multivariable function defined on domain $D \subset R^3$. Which of the following is(are) correct?

☐ f is a scalar valued multivariable function.

☐ The rate of change of the function f at (0,1,1) with respect to x is 1.

☐ The rate of change of the function f at (2,2,2) with respect to x is same as the rate of change of the function f at (2,2,2) with respect to z.

☐ The rate of change of the function f at (1,3,5) with respect to y is -9.

No, the answer is incorrect.
Score: 0
Accepted Answers:
f is a scalar valued multivariable function.
The rate of change of the function f at (2,2,2) with respect to x is same as the rate of change of the function f at (2,2,2) with respect to z.
The rate of change of the function f at (1,3,5) with respect to y is -9.
1 point
Suppose $f: D \longrightarrow R$ is a multivariable function defined on domain $D \subset R^2$ and also satisfying the following conditions:

(i) $\frac{\partial f}{\partial x} = 6x + 4xy + 3y$
(ii) $\frac{\partial f}{\partial y} = 2x^2 + 3x - 3y$

Which of the following functions can not be f?

☐ $f(x, y) = 3x^2 + 2x^2y + 3xy - \frac{3}{2}y^2$

☐ $f(x, y) = 3x^2 + 2x^2y - 3xy - \frac{3}{2}y^2$

☐ $f(x, y) = 3x^2 - 3xy - 2x^2y + 3y^2$

☐ $f(x, y) = 3x^2 + 3y^2 + 3xy$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y) = 3x^2 + 2x^2y - 3xy - \frac{3}{2}y^2$
$f(x, y) = 3x^2 - 3xy - 2x^2y + 3y^2$
$f(x, y) = 3x^2 + 3y^2 + 3xy$

Let $f: R^2 \rightarrow R^2$ be given by $f(x, y) = (P(x, y), Q(x, y))$, where $P(x, y) = 3x^2y, Q(x, y) = 5x + y^3$. Consider a matrix $A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$. If the determinant of A is $axy^3 - bx^2$ where $a, b \in R$, then find the value of $a - b$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Let $f: R^2 \rightarrow R^2$ be given by $f(x, y) = (P(x, y), Q(x, y))$, where $P(x, y) = 6x + y^2, Q(x, y) = 3x^2 + 2y$. Consider a matrix $A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$. If $det(A) = 0$, then find the value of xy

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

Let $f: R^2 \rightarrow R$ be defined as $f(x, y) = e^{4(x+y)} sin(7x - 4y)$. If $4\frac{\partial f}{\partial x} + 7\frac{\partial f}{\partial y} = k.f(x, y)$ where $k \in R$, then find the value of k.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 44
1 point

Check Answers

Your score is: 0/9